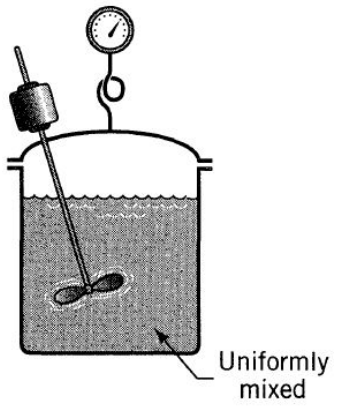
LMAPR 2430: Exercise session 1

BATCH reactors



$$\frac{dx_A}{d\theta} = \frac{V}{N_{A0}} r_A\left(x_A, T\right)$$

$$m_t c_p \frac{dT}{d\theta} = V\left(-\Delta H\right) r_A\left(x_A, T\right) + qA_k$$

1 A gas-phase decomposition $A \longrightarrow R + S$ is carried out with initial conditions of $T_0 = 300$ K, $p_t = 5$ bar and $V_0 = 0.5$ m^3. The heat of reaction is -6280 kJ/kmol and the heat capacity of A, R and S are 125.6, 104.7 and 83.7 kJ/kmol.K respectively. The rate coefficient is

$$k = 10^{14} \exp\left(\frac{-10\,000}{T}\right) \qquad h^{-1}$$

1. For isothermal conditions:
   (a) Calculate the required time to obtain 85% conversion of A.
   (b) Draw the conversion-time profile.
   (c) Determine the heat exchange rate required to maintain isothermal conditions.
2. For adiabatic conditions, the temperature is no longer constant and becomes a function of conversion.
   (a) Obtain the function to integrate to calculate the required time to obtain 85% conversion of A.
   (b) Using MATLAB and an adequate numerical method, integrate this function.
   (c) Draw the conversion vs. time profile and compare with the case of isothermal conditions. Explain.

2 In a batch reactor with a volume $V = 5$ m^3, an exothermic reaction $A \longrightarrow P$ is carried out in the liquid phase. The rate equation is

$$r_A = kC_A$$

with

$$k = 4 \times 10^6 \exp\left(\frac{-7900}{T}\right) \qquad s^{-1}$$

The initial temperature T_0 of the reaction mixture is 20°C and the maximum allowable reaction temperature is 95°C. The reactor contains a heat exchanger with area $A_k = 3.3$ m^2 and it can be operated with steam ($T_r = 120$°C and $U = 1360$ W/m^2.K) or with cooling water ($T_r = 15$°C and $U = 1180$ W/m^2.K). The times required for filling and emptying the reactor are 10 and 15 min respectively.
The desired conversion is $x_{Af} \geq 0.9$. Determine the complete reaction cycle times (heat, reaction, cooling), along with the steam and water consumption rates for the following policies of operation:

1. Heat to 95°C, let the reaction proceed isothermally until $x_A = 0.9$ occurs, cool down to 45°C.
2. Preheat to 55°C, let the reaction proceed adiabatically, start cooling when either $T = 95$°C or $x_A = 0.9$ occurs, and cool down to 45°C. For this question, numerical integration is required.

Additional physicochemical data:

ΔH	-1670	kJ/kg
ρc_p	4.2×10^6	$J/(m^3.K)$
M_A	100	$kg/kmol$
C_{A0}	1	$kmol/m^3$

3 For a reversible exothermic reaction $\mathrm{A} \rightleftharpoons \mathrm{Q} + \mathrm{S}$, we want to chose the optimal operating temperature variation with time. This kind of operation is feasible in practice by using automatic control techniques. First, discuss qualitatively the optimum temperature profile to select. Then, find the equation of the optimal temperature profile $T_{opt}(x_A)$ to be imposed. The condition to fulfil is to impose at each point the temperature that maximizes the reaction rate.
For this case:

$$\mathrm{A} \underset{2}{\overset{1}{\rightleftharpoons}} \mathrm{Q} + \mathrm{S}$$

$$r_A = A_1 \exp\left(\frac{-E_1}{RT}\right) C_A - A_2 \exp\left(\frac{-E_2}{RT}\right) C_Q C_S.$$

4

The first-order liquid phase reversible and exothermic reaction

$$A \underset{2}{\overset{1}{\rightleftharpoons}} R$$

is carried out in a well stirred batch reactor. The following rate parameters are given:

$$A_1 = 7 \ \ s^{-1} \qquad E_1 = 41868 \ \ J/mol$$
$$A_2 = 5000 \ \ s^{-1} \qquad E_2 = 83736 \ \ J/mol$$

Temperature is controlled using a jacket, as shown in Figure 1, and the maximum allowable temperature in the reactor is 800 K. The reactant A is fed with a solvent S (inert) in a 1:1 molar ratio.

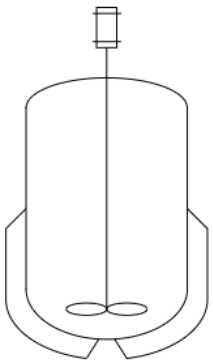


Figure 1.1 – Batch reactor

A conversion of 0.8 is desired and you are asked to optimize the operating conditions to minimize the time required. The following additional data are available:

ρ_L	1000	kg/m^3
y_{A0}	0.5	
M_A	100	kg/kmol
M_S	20	kg/kmol
c_p	4.187	kJ/kg.K

1. Determine the optimal temperature profile as a function of conversion, in the absence of any other constraint, in order to minimize the reaction time.
2. Modify the temperature profile obtained at previous point, knowing that the maximum allowable temperature in the reactor is 800 K, and plot it vs. conversion.
3. Determine the evolution of conversion and temperature with time by integrating the continuity equation. In order to do so, implement a numerical routine such as Runge-Kutta, Heun or Euler method. Plot the profiles of conversion and temperature vs. time and determine the time required to reach the requested conversion.
4. Determine the heat exchange rate per unit reactor volume required (kJ/m$_r^3$.s) and plot it vs. conversion.

Additional exercises

5 A daily production of 50 000 kg (50 tons metric) of ethyl acetate is to be produced in a batch reactor from ethanol and acetic acid:

$$\underset{\text{(A)}}{C_2H_5OH} + \underset{\text{(B)}}{CH_3COOH} \rightleftharpoons \underset{\text{(R)}}{CH_3COOC_2H_5} + \underset{\text{(S)}}{H_2O}$$

alcohol acid ethyl acetate

The reaction rate in the liquid phase is given by:

$$r_A = k\left(C_A C_B - \frac{C_R C_S}{K}\right)$$

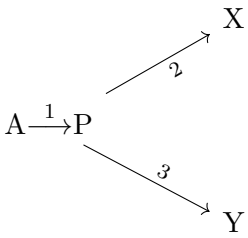
At 100°C,

$$k = 7.23 \times 10^{-6} \ \ m^3/(kmol.s)$$
$$K = 2.93$$

X_B

A feed of 23 percent by weight of acid, 46 percent of alcohol and no ester is to be used, with a 35 percent conversion of acid. The density is essentially constant at 1020 kg/m^3. The reactor will be operated 24h per day, and the time for filling, emptying, and the like, is 1h total for reactors in the contemplated size range. What reactor volume is required?

6 A desired product P is made according to the following reactions scheme:



$$k_i = A_i e^{-E_i/RT}$$

Discuss qualitatively the optimum temperature profile for the two cases:

1. $E_2 > E_3 > E_1$,
2. $E_2 > E_1 > E_3$.

Describe your reasoning *carefully*.

7 An endothermic third-order reaction $3A \longrightarrow 2B + C$ is carried out in a batch reactor. The reaction mixture is heated to 400°C. The reaction then proceeds adiabatically. During the heating period, 10 mol-% of A is converted. From this instant on, what is the time required to reach a conversion of 70%?
The kinetic constant is given by:

$$\ln k = \frac{-10\,000}{RT} + 5 \qquad k \text{ in } m^6/(kmol_A^2.s)$$

Additional physicochemical data:

ΔH	104 670	$kJ/kmol$
V	1	m^3 (constant)
c_{pm}	2.47	$kJ/(kg.K)$
m_t	950	kg
N_{A0}	10.2	$kmol$

REMINDER

X_A = reaction rate (e.g. 85%)
r_A = reaction speed (e.g.

Write the heat and conservation equation for the batch reactor

$$A_{in} - A_{out} = A_{reacted} - A_{accumulated}$$

$A_{in} = 0$ and $A_{out} = 0$ because we are in a batch

EXERCISE 1

1. For isothermal conditions $\Rightarrow k = cst$

(a) $$\frac{dN_A}{dt} = (-r_A) \cdot V \quad (1)$$

$\left[\frac{mol}{s}\right]$ $\left[\frac{mol}{m^3 \cdot s}\right]$ $[m^3]$

et $N_A = N_{A_0} \cdot (1 - X_A) = N_{A_0} - N_{A_0} \cdot X_A$

$\Leftrightarrow dN_A = - N_{A_0} \cdot dX_A \quad (2)$

(2) → (1) :

$$\frac{- N_{A_0} \, dX_A}{dt} = (-r_A) \cdot V$$

$$\Leftrightarrow \frac{dX_A}{dt} = r_A \cdot \frac{V}{N_{A_0}}$$

→ general equation for batch (mass balance)

$$\Leftrightarrow \int dX_A \cdot \frac{N_{A_0}}{V} \cdot \frac{1}{r_A} = \int dt$$

$$\Leftrightarrow t = N_{A_0} \cdot \int_{X_{A_0}}^{X_{A_f}} \frac{1}{V \cdot r_A} \, dX_A$$

$$= N_{A_0} \cdot \int \frac{1}{V \cdot r_A(X_A)} \, dX_A$$

on veut exprimer r_A en fonction de X_A :

$r_A = k \cdot C_A^n$ on est dans une équation d'ordre 1 $\Rightarrow n = 1$

$\Leftrightarrow r_A = k \cdot C_A$ par def $C_A = \frac{N_A}{V}$ $\left[\frac{mol}{m^3}\right]$

$$= k \cdot \frac{N_A}{V}$$

$$= \frac{k}{V} \cdot N_{A_0} \cdot (1 - X_A) \quad (*)$$

$$= N_{A_0} \cdot \int \frac{1}{V} \cdot \frac{V}{k \cdot N_{A_0} \cdot (1 - X_A)} \, dX_A$$

$$= N_{A_0} \cdot \int \frac{1}{k \cdot N_{A_0} \cdot (1 - X_A)} \cdot dX_A$$

$$= \frac{N_{A_0}}{k \cdot N_{A_0}} \int \frac{1}{1 - X_A} \, dX_A \qquad \text{on pose} \quad 1 - X_A = u$$

$$\Leftrightarrow \quad -dX_A = du$$

$$\Leftrightarrow \quad dX_A = -du$$

$$= \frac{1}{k} \int \frac{1}{u} \, -du$$

$$= \frac{-1}{k} \cdot \ln(u)$$

$$t = \frac{-1}{k} \cdot \left[\ln(1 - X_A) \right]_{X_{A_0} = 0}^{X_{Af}}$$

$$t = \frac{-1}{k} \cdot \ln(1 - X_A)$$ tjrs valable pour cas isothermes

How much time do we need to get 85% of conversion ?

* $k = 10^{14} \exp\left(\frac{-10\,000}{T}\right) \quad h^{-1}$ où $T_0 = 300\ K$

$$\Leftrightarrow \quad k = 10^{14} \cdot \exp\left(\frac{-10\,000}{300}\right)$$

$$= 0,334\ h^{-1}$$

* $X_A = 0,85$

$$\Longrightarrow \quad t = \frac{-1}{0,334} \cdot \ln(1 - 0,85)$$

$$= 5,68\ h$$

$\cdot 60$

$$= 5\ h\ 40,8\ min$$

$\cdot 60$

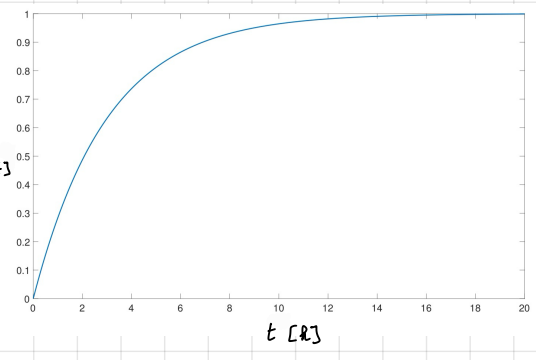
$$= 5\ h\ 40\ min\ 48\ s$$

(b) Conversion time profile ?

isoler X_A dans cette équation $t = \frac{-1}{k} \cdot \ln(1 - X_A)$

$$\Leftrightarrow \quad -tk = \ln(1 - X_A)$$

$$\Leftrightarrow \quad e^{-tk} = 1 - X_A$$



$$\Leftrightarrow \boxed{X_A = 1 - e^{-tk}} \quad (**)$$

(c) Heat exchange rate to maintain isothermal conditions

Heat balance : accumulation = in - out + generation - consumption

$$m \cdot C_p \cdot \frac{dT}{dt} = (-\Delta H)\, r_A \cdot V + Q$$

$\frac{dT}{dt} = 0$ because we want isothermal

$$\Leftrightarrow Q = \Delta H \cdot r_A \cdot V \quad \text{(from (*))}$$

$$\Leftrightarrow Q = \Delta H \cdot \frac{k}{V} \cdot N_{A_0} \cdot (1 - X_A) \cdot V \quad \text{(from (**))}$$

$$= \Delta H \cdot k \cdot N_{A_0} \cdot (1 - 1 + e^{-tk})$$

$$= \Delta H \cdot k \cdot N_{A_0} \cdot e^{-tk}$$

$$= \Delta H \cdot k \cdot \frac{P_0 \cdot V_0}{R \cdot T_0} \cdot e^{-tk}$$

par la loi des gaz parfaits : $PV = n \cdot R \cdot T$

$$\Leftrightarrow N_{A_0} = \frac{P_0 \cdot V_0}{R \cdot T_0}$$

$$= -6\,280 \cdot 0{,}334 \cdot \frac{5 \cdot 10^5 \cdot 0{,}5}{8{,}314 \cdot 300} \cdot e^{-0{,}334t}$$ ([Pa], [m³], [K])

$$= -210\,234{,}43 \cdot e^{-0{,}334t} \quad [J]$$

Selon les unités utilisées, la valeur de R varie :

Unités utilisées	Expression de la loi	Valeur de R	Unités de R
Système international (SI)	$PV = nRT$	8,314462618	J·mol⁻¹·K⁻¹
En L·bar·mol⁻¹·K⁻¹	$PV = nRT$	0,08314	L·bar·mol⁻¹·K⁻¹
En L·atm·mol⁻¹·K⁻¹	$PV = nRT$	0,08206	L·atm·mol⁻¹·K⁻¹
En cal·mol⁻¹·K⁻¹	$PV = nRT$	1,987	cal·mol⁻¹·K⁻¹

→ So, over the entire reaction the total amount of energy is :

$$E = \int_{t_0}^{t_f} -210\,234{,}43 \cdot e^{-0{,}334t}\, dt$$

$$= -210\,234{,}43 \cdot \frac{1}{-0{,}334} \cdot \left[e^{-0{,}334t}\right]_{t_0}^{t_f}$$

$$= -210\,234{,}43 \cdot \frac{1}{-0{,}334} \cdot \left(e^{-0{,}334 \cdot 0} - e^{-0{,}334 \cdot 5{,}68}\right)$$

$$= -210\,234{,}43 \cdot \frac{1}{-0{,}334} \cdot \left(1 - e^{-0{,}334 \cdot 5{,}68}\right)$$

$$= -535\,090{,}7317 \text{ J}$$

$$= -535{,}09 \text{ kJ}$$

2. For adiabatique conditions ⇒ k ≠ constant

(a) Obtain the function to integrate to calculate the required time to obtain 85% conversion of A.

$t = f(x_A)$ but now $k = f(T)$ ...

but the temperature is also a function of the reaction since it is exothermic (the t° varies along with the progress of the reaction)

⟶ $T = f(x_A)$

Starting from the energy balance:

$$m \cdot C_p \cdot \frac{dT}{dt} = -\Delta H \cdot r_A \cdot V + \underbrace{Q}_{=0}$$

car adiabatique

$$\Leftrightarrow \quad dT = \frac{-\Delta H \cdot r_A \cdot V}{m \cdot C_p} \cdot dt \qquad \text{or, je sais que } r_A = \frac{k}{V} \cdot N_{A_0} \cdot (1 - x_A)$$

$$\Leftrightarrow \quad dT = \frac{-\Delta H \cdot \frac{k}{V} \cdot N_{A_0} \cdot (1-x_A) \cdot V}{m \cdot C_p} \, dt$$

$$\Leftrightarrow \quad dT = \frac{-\Delta H \cdot k \cdot N_{A_0} \cdot (1-x_A)}{m \cdot C_p} \, dt$$

Careful because here k is not constant, so this equation cannot be solved individually... In adiabatic operation, the continuity and energy equations have to be solved together:

$$\begin{cases} \dfrac{dx_A}{dt} = \dfrac{V}{N_{A_0}} r_A(x_A, T) & \text{(mass balance)} \\ m_t \, c_p \dfrac{dT}{dt} = r_A(x_A, T)\, V\, (-\Delta H) & \text{(energy balance)} \end{cases} \quad (2)$$

(= continuity equation)

$$\Leftrightarrow \quad r_A = \frac{dx_A}{dt} \cdot \frac{N_{A_0}}{V} \quad (1)$$

(1) → (2)

$$m \cdot c_p \cdot \frac{dT}{dt} = \frac{dx_A}{dt} \cdot \frac{N_{A_0}}{V} \cdot V \cdot (-\Delta H)$$

$$\Leftrightarrow \quad m \cdot c_p \cdot dT = (-\Delta H) \cdot N_{A_0} \, dx_A$$

⚠ c_p is function of x_A

↳ $m_t \cdot c_p = N_t \cdot (y_A \cdot C_{p_A} + y_R \cdot C_{p_R} + y_S \cdot C_{p_S})$

$[kg] \cdot \left[\frac{kJ}{kg \cdot K}\right] = [mol] \cdot [-] \left[\frac{kJ}{kmol \cdot K}\right]$

$$= N_t \cdot \left(\frac{N_A}{N_t} \cdot C_{p_A} + \frac{N_R}{N_t} \cdot C_{p_R} + \frac{N_S}{N_t} \cdot C_{p_S} \right)$$

reaction:

$A \longrightarrow R + S$

$$= N_{A_0} \cdot (1 - x_A) \cdot C_{p_A} + N_{A_0} \cdot x_A \cdot C_{p_R} + N_{A_0} \cdot x_A \cdot C_{p_S}$$

⟶ NOTE : variation maximale de la T d'un batch adiabatique est lorsque $x_A = 1$ et $x_{A_0} = 0$.

$$= N_{A_0} \cdot \left[(1 - X_A) \cdot C_{pA} + X_A \cdot (C_{pR} + C_{ps}) \right]$$

$$= N_{A_0} \cdot \left[C_{pA} - C_{pA} \cdot X_A + X_A \cdot (C_{pR} + C_{ps}) \right]$$

$$= N_{A_0} \cdot \left[C_{pA} + X_A \cdot (C_{pR} + C_{ps} - C_{pA}) \right]$$

$$\Leftrightarrow \quad N_{A_0} \cdot \left[C_{pA} + X_A \cdot (C_{pR} + C_{ps} - C_{pA}) \right] dT = (-\Delta H) \cdot N_{A_0} \, dX_A$$

$$\Leftrightarrow \int dT = \int (-\Delta H) \frac{dX_A}{C_{pA} + X_A \cdot (C_{pR} + C_{ps} - C_{pA})}$$

$$\Leftrightarrow \quad T - T_0 = (-\Delta H) \cdot \ln \left[C_{pA} + X_A \cdot (C_{pR} + C_{ps} - C_{pA}) \right] \cdot \frac{1}{C_{pR} + C_{ps} - C_{pA}}$$

$$\Leftrightarrow \quad T = T_0 + \frac{(-\Delta H)}{C_{pR} + C_{ps} - C_{pA}} \cdot \ln \left[C_{pA} + X_A \cdot (C_{pR} + C_{ps} - C_{pA}) \right]$$

$$= 300 + \frac{-(-6280)}{104,7 + 83,7 - 125,6} \cdot \ln \left[125,6 + X_A \cdot (104,7 + 83,7 - 125,6) \right]$$

$$\Leftrightarrow \quad T = 300 + 100 \ln (125,6 + 62,8 X_A)$$

Now that we have determined the evolution of temperature with conversion, we can reuse the continuity equation to determine the time required to reach a desired conversion :

$$\frac{dX_A}{dt} = \frac{V}{N_{A_0}} \cdot r_A$$

$$\Leftrightarrow \quad dX_A \cdot \frac{N_{A_0}}{V} \cdot \frac{1}{r_A} = dt$$

$$\Leftrightarrow \quad dt = \frac{N_{A_0}}{V} \cdot \frac{1}{k \cdot C_A} dX_A$$

$$= \frac{N_{A_0}}{V} \cdot \frac{1}{k \cdot \frac{N_A}{V}} dX_A$$

$$= \frac{N_{A_0}}{V} \cdot \frac{1}{k} \cdot \frac{V}{N_{A_0} (1 - X_A)} dX_A$$

$$\Leftrightarrow \quad \int dt = \int \frac{1}{k} \cdot \frac{1}{(1 - X_A)} dX_A$$

$$\Leftrightarrow \quad t_g (X_{Ag} = 0,85) = \int_0^{X_{Ag} = 0,85} \frac{1}{10^{14} \exp \left(\frac{-10\,000}{300 + 100 \ln (125,6 + 62,8 X_A)} \right)} \cdot \frac{1}{(1 - X_A)} dX_A$$

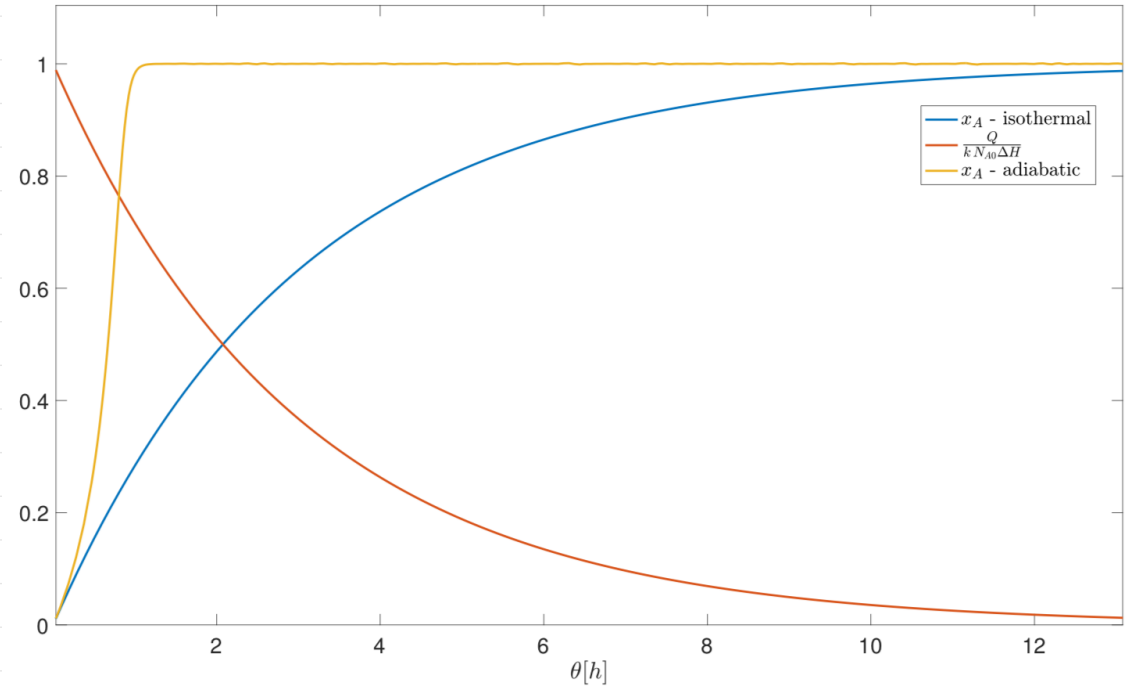
$\longrightarrow$ This integral must be computed numerically

(b) After numerical resolution, we obtain : (Runge - Kutta method)

$t_g\ (X_{Ag} = 0{,}85) = 0{,}86\ h \approx 51\ min$

Indeed, under a diabetic conditions, the exothermic reaction drives its own kinetic (T increases and k increases), reaching faster the desired conversion.

(c) Graph of the conversion - time profile :



Interpretations:

Blue curve – x_A under isothermal conditions: This is the first-order batch profile. Since the temperature is kept constant, the rate constant is fixed, so the conversion increases smoothly and gradually with time. There is no thermal acceleration.

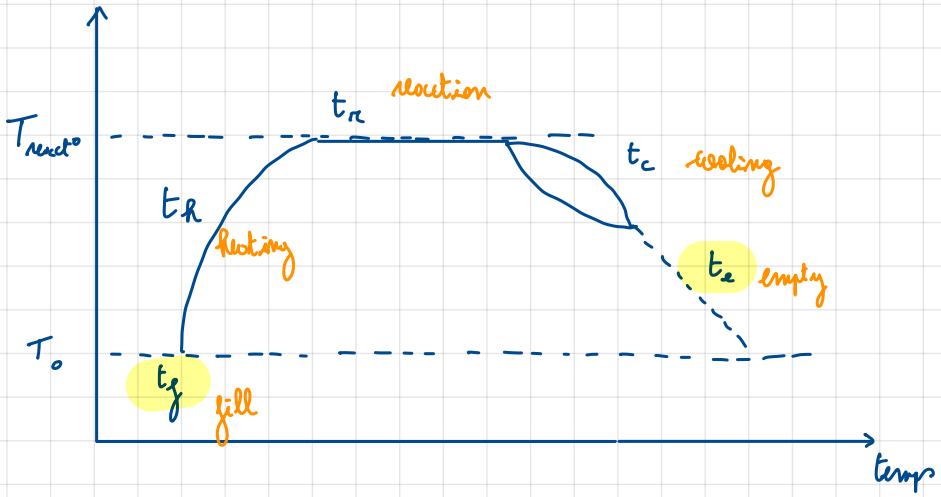
Yellow curve – x_A under adiabatic conditions: In the adiabatic case all the reaction heat stays inside the system. Because the reaction is exothermic, the temperature rises as A is consumed. The rate constant depends exponentially on temperature, so even a moderate temperature increase produces a very large increase in reaction rate. As a result, the conversion rises extremely rapidly. This explains why the yellow curve shoots up sharply and reaches almost complete conversion much earlier than in the isothermal case.

Red curve – the fraction of A remaining: The red curve shows the amount of reactant left, 1 - x_A. Under isothermal conditions it decays exponentially in time, consistent with (*). Since the heat released is proportional to the amount of A that has reacted, this curve is also related to the fraction of total reaction heat released relative to the maximum possible heat at full conversion.

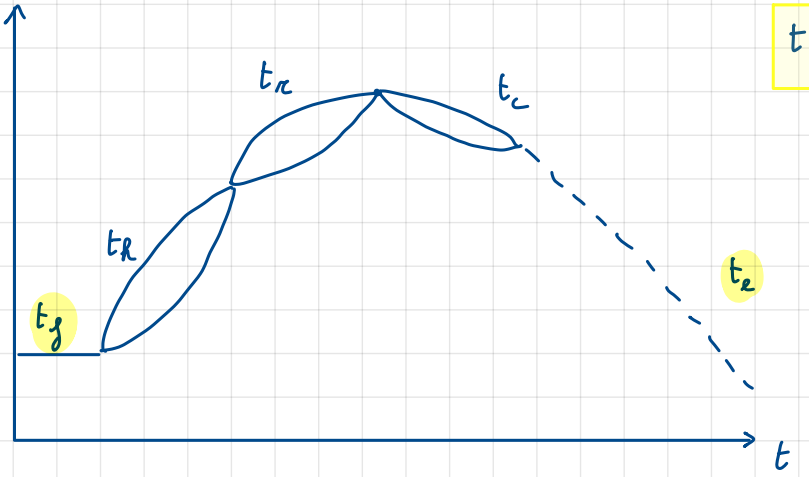
$$(*) = 1 - X_A = \exp(-kt) = \frac{Q}{k \cdot N_{A0} \cdot \Delta H}$$

EXERCISE 2

EXOTHERMIQUE

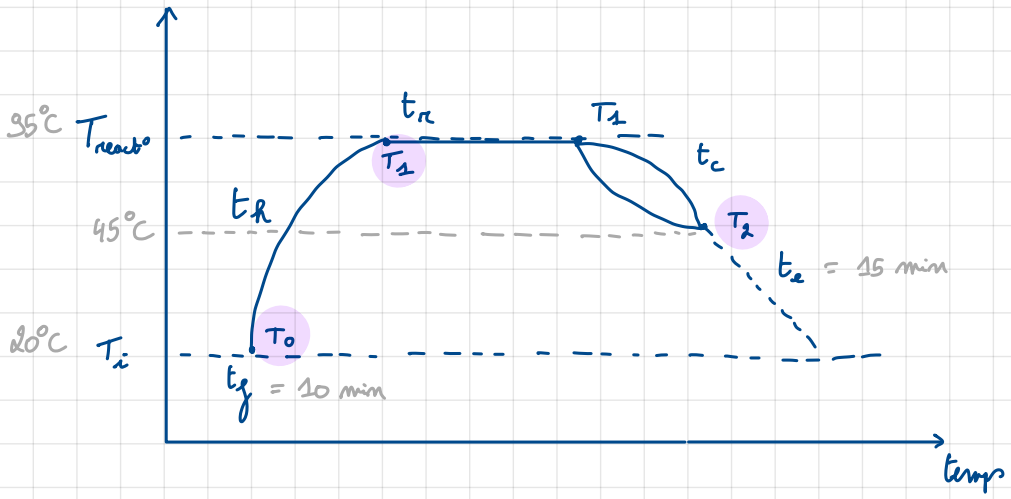


ADIABATIQUE



$$t = t_g + t_R + t_r + t_c + t_e$$

1. ISOTHERM



On nous demande de calculer le temps :

On sait que pour remplir et pour vider le batch c'est 10 et 15 min (donné dans l'énoncé). Il nous faut alors calculer le temps de chauffe, de refroidissement et de réaction.

- **Temps de chauffe : t_R**

$$m_t c_p \frac{dT}{d\theta} = V(-\Delta H) r_A(x_A, T) + qA_k$$

$\frac{dT}{d\theta}$: temps ; $V(-\Delta H) r_A(x_A, T)$: chaleur dég. par la réaction ; qA_k : aire échange de chaleur ; $V \cdot dT$

$= 0$ pas de réaction, on chauffe juste les parois du batch avec de la vapeur d'eau

$$\Leftrightarrow m_t \, c_p \, \frac{dT}{dt} = U \cdot \Delta T \cdot A_R$$

↳ ΔT : ce de T° entre le milieu extérieur et le milieu intérieur

$$\Leftrightarrow m_t \cdot c_p \cdot dT = U \cdot (T_{steam} - T_{int}) \cdot A_R \cdot dt$$

$$\Leftrightarrow m_t \cdot c_p \cdot \frac{1}{T_{steam} - T_{int}} \cdot dT = U \cdot A_R \cdot dt$$

T_{steam} = 120°C = température de la vapeur d'eau autour de mon batch pour chauffer

$$[kg] \left[\frac{J}{kg \cdot K}\right] \cdot \left[\frac{1}{K}\right] [K]$$

$$\left[\frac{J}{K}\right]$$

$$\left[\frac{J}{K \cdot m^3}\right] (m^3)$$

$$\Leftrightarrow \rho \cdot c_p \cdot V \cdot \frac{1}{120 - T_{int}} \, dT = 1360 \cdot 3{,}3 \, dt$$

$$\Leftrightarrow \int_{20°C}^{35°C} 4{,}2 \cdot 10^6 \cdot 5 \cdot \frac{1}{120 - T_{int}} \, dT = \int_{t_i}^{t_f} 1360 \cdot 3{,}3 \, dt$$

$$\Leftrightarrow 21 \cdot 10^6 \cdot \left[-\ln(120 - T_{int}) \right]_{20°C}^{35°C} = 4488 \cdot t$$

$\Leftrightarrow \quad 21.10^6 . \left[-\ln(120-35) - (-\ln(120-20)) \right] = 4488.t$

$\Leftrightarrow \quad t_h = \dfrac{21.10^6}{4488} . \ln\left(\dfrac{(120-20)}{(120-35)}\right) \longrightarrow t_h = \dfrac{V(\rho c_p)}{U_{steam} A_k} \ln\left(\dfrac{T_{steam}-T_0}{T_{steam}-T_1}\right)$

$= 6,4867.10^3$ s

$\downarrow \div 60$

$= 108,11$ min

$= 1,8$ h

$\hookrightarrow .60 \quad = \boxed{1\text{ h }48\text{ min}}$

- **Temps de refroidissement : t_c**

t° de l'air autour de mon batch pour refroidir

$t_c = \dfrac{V.\rho.c_p}{U_{cold}.A_k} . \ln\left(\dfrac{T_{cold}-35}{T_{cold}-45}\right) \longrightarrow t_c = \dfrac{V(\rho c_p)}{U_{cold} A_k} \ln\left(\dfrac{T_{cold}-T_1}{T_{cold}-T_2}\right)$

$= \dfrac{5.4,2.10^6}{1180.3,3} . \ln\left(\dfrac{15-35}{15-45}\right)$

$= 5,2895.10^3$ s

$= 8,8159.10$ min

$= 1,46$ h $= \boxed{1\text{ h }28}$

- **Temps de réaction : t_R**

$t_R = \dfrac{-1}{k} \ln(1-x_a)$ où $k = k_0 . \exp\left(\dfrac{-E_a}{RT}\right)$

$= \dfrac{-1}{0,0019} . \ln(1-0,9)$ $= 4.10^6 . \exp\left(\dfrac{-7300}{35+273}\right)$ via énoncé

$= 0,0019 \ s^{-1}$

$= 1,2010.10^3$ s

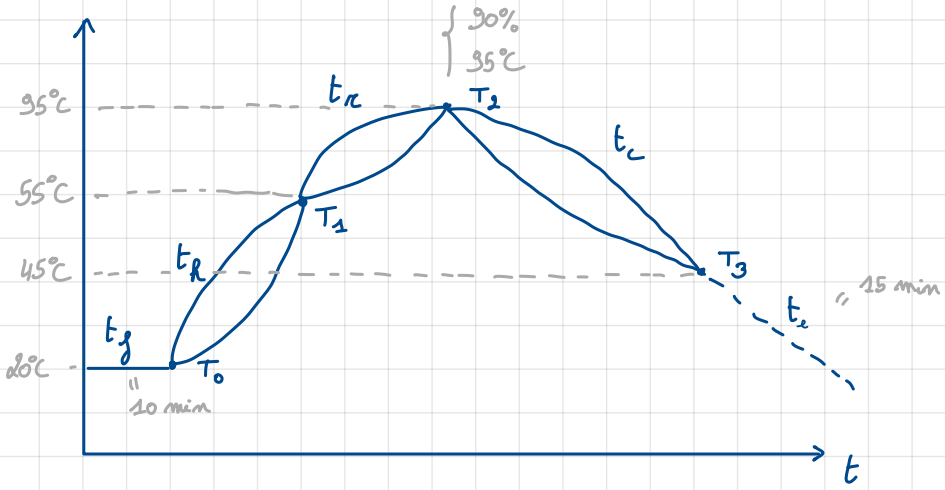
$= 2,0016.10$ min $= \boxed{20\text{ min}}$

- **Conclusion**

$\longrightarrow$ 15 min + 10 min + 1 h 48 min + 1 h 28 min + 20 min

$= 241$ min ≈ 4 h

2. ADIABATIC



- **Temps de réaction : t_R**
  - On calcule la température qui est fonction du taux de conversion xA
  - Grâce à la température on peut calculer k
  - Je réinjecte k dans l'équation de mon temps

(1) $$\frac{dx_A}{d\theta} = \frac{V}{N_{A0}} r_A(x_A, T)$$

(2) $$m_t c_p \frac{dT}{d\theta} = V(-\Delta H) r_A(x_A, T) + q A_k$$

(température ; $V \cdot dT$; temps ; chaleur dég. par la réaction ; air échange de chaleur)

$$m_T \cdot C_p \cdot \frac{dT}{dt} = V \cdot (-\Delta H) \cdot r_A + \underbrace{q \cdot A_k}_{=0 \text{ car adiabatique}}$$

on injecte $\frac{dx_A}{d\theta} = \frac{V}{N_{A0}} r_A(x_A, T)$

$$\Leftrightarrow m_T \cdot C_p \cdot \frac{dT}{dt} = V \cdot (-\Delta H) \cdot \frac{dx_A}{dt} \cdot \frac{N_{A0}}{V} \quad \text{avec } C_{A0} = \frac{N_{A0}}{V}$$

$$\Leftrightarrow \int \rho \cdot C_p \cdot dT = \int -\Delta H \cdot dx_A \cdot C_{A0}$$

car on ne connait pas N_{A0} mais C_{A0} nous est donné

$$\Leftrightarrow [T]_{SS}^{T} = \frac{-\Delta H}{\rho \cdot C_p} \cdot C_{A0} \cdot X_A$$

$$\Leftrightarrow T - (55+273) = \frac{-\Delta H}{\rho \cdot C_p} \cdot C_{A0} \cdot X_A$$

$$\Leftrightarrow T = \frac{-(-1670 \cdot 10^3)}{4{,}2 \cdot 10^6} \cdot 10^3 \; 0{,}1 \cdot 0{,}9 + \underbrace{(55+273)}_{T_1}$$

$$\frac{[J/kg]}{[J/m^3 \cdot K]} \cdot \left[\frac{mol}{m^3}\right] \cdot \left[\frac{kg}{mol}\right] \cdot [-] = \frac{J}{kg} \cdot \frac{m^3 K}{J} \cdot \frac{mol}{m^3} \cdot \frac{kg}{mol} = [K]$$

$$= 363{,}78\ K = 91^\circ C$$

On nous dit dans l'énoncé qu'on commence à cool down **soit** quand la température atteint 95°C **soit** quand le taux de conversion atteint 0,9. On calcule donc la température à un taux de conversion de 0,9 et on voit quelle température on obtient. Si on obtient une température supérieure à 95°C on oublie et on prend 95°C comme référence, par contre si on obtient une température inférieure à 95°C on prend cette nouvelle température comme référence.

On a une température inférieure à 95°C donc on garde cette température en référence. Mais attention, cette température va toujours évoluer dans le réacteur... on ne peut donc pas la prendre pour acquise.

L'expression de T en fonction de x_1 est donc :

$$T = \frac{-\Delta H}{\rho . C_p} . C_{A_0} . X_A + T_1$$

Maintenant qu'on connaît la température de notre réaction, on peut calculer l'expression de k:

$$k = k_0 . \exp\left(\frac{-E_a}{RT}\right) = 4.10^6 . \exp\left(\frac{-7300}{T}\right)$$

Maintenant qu'on a k on peut réinjecter dans l'équation du temps et trouver et « temps réaction »:

$$\frac{dN_A}{dt} = (-r_A) . V \quad \text{où} \quad r_A = k . C_A$$

$$= -k . \frac{N_A}{V} . V \qquad = k . \frac{N_A}{V}$$

$$= -k . \frac{N_{A_0} . (1 - X_A)}{V} . V$$

$$\Leftrightarrow dN_A = -k . N_{A_0} . (1 - X_A) . dt \qquad N_A = N_{A_0} . (1 - X_A)$$

$$\Leftrightarrow -N_{A_0} dX_A = -k N_{A_0} . (1 - X_A) dt \qquad \Leftrightarrow N_A = N_{A_0} - N_{A_0} X_A$$

$$\Leftrightarrow \int dt = \int \frac{1}{k} . \frac{1}{(1 - X_A)} dX_A \qquad \Rightarrow dN_A = -N_{A_0} dX_A$$

$$\Leftrightarrow t = \int \frac{1}{k} . \frac{1}{(1 - X_A)} dX_A$$

$$= \int_0^{0,9} \frac{1}{4.10^6 \exp\left(\frac{-7300}{\frac{-\Delta H}{\rho . C_p} . C_{A_0} . X_A + T_1}\right)} . \frac{1}{(1 - X_A)} dX_A$$

→ à résoudre numériquement

$= 4\,065\ s \approx$ **68 min**

ou en partant de :

$$\frac{dX_A}{dt} = r_A . \frac{V}{N_{A_0}}$$

$$\Leftrightarrow \frac{1}{r_A} . \frac{N_{A_0}}{V} dX_A = dt$$

$$\Leftrightarrow \frac{1}{k . C_A} . C_{A_0} . dX_A = dt$$

$$\Leftrightarrow \frac{1}{k . C_{A_0} . (1 - X_A)} . C_{A_0} . dX_A = dt$$

$$\Leftrightarrow \frac{1}{k . (1 - X_A)} dX_A = dt$$

- **Temps de heating : t_k**

Starting from the energy balance :

$$m_T . C_p . \frac{dT}{dt} = \underbrace{V . (-\Delta H) . r_A}_{=0} + q . A_k$$

car adiab.
→ no heat released by the reaction

delta entre T_{steam} et T_{int}.

$$\Leftrightarrow \rho . V . C_p . dT = U . \Delta T . A_R . dt$$

$$\Leftrightarrow \int_{20°C}^{55°C} \rho . V . C_p . \frac{1}{(120 - T_{int})} dT = \int_{t_i}^{t_f} U . A_R \, dt$$

$$\Leftrightarrow \rho . C_p . V . \left[- \ln (120 - T_{int}) \right]_{20°C}^{55°C} = U . A_R . t$$

$$\Leftrightarrow t = \frac{\rho . C_p . V}{U_{steam} . A_R} . \ln \left(\frac{120 - 20}{120 - 55} \right)$$

$$= \frac{4,2 . 10^6 . 5}{1360 . 3,3} . \ln \left(\frac{120 - 20}{120 - 55} \right)$$

$$= 2015,69 \text{ s} = 33,59 \text{ min} \approx \boxed{34 \text{ min}}$$

- **Temps de cooling : t_c**

$$t = \frac{\rho . C_p . V}{U_{cold} . A_R} . \ln \left(\frac{T_{cold} - T_2}{T_{cold} - T_3} \right)$$

⚠ à bien reprendre la t° calculée vu que qd $x = 0,9$ $T < 95°C$

$$= \frac{4,2 . 10^6 . 5}{1180 . 3,3} . \ln \left(\frac{15 - 91}{15 - 45} \right)$$

$$= 5012,9 \text{ s} = 83,5 \text{ min} \approx \boxed{83 \text{ min}}$$

- **Conclusion**

$\longrightarrow$ 15 min + 10 min + 68 min + 34 min + 83 min

$= 210 \text{ min} = 3 \text{ h } 30$

EXERCICE 3

r_A = rate ("avancement de la réaction") (kinetic properties)
↳ positive for forward

X_A = taux de conversion $X_A = 0 \rightarrow$ no reaction yet $X_A = 1 \rightarrow$ complete conversion

D'abord augmenter la température pour casser l'énergie d'activation et « drive kinetics at low conversion ». Ensuite comme c'est une réaction exothermique, pour favoriser la formation de produits il faut justement diminuer la température.

Find the equation of the optimal temperature that maximizes the reaction rate

$\longrightarrow$ optimization problem : $\frac{dr_1}{dT} = 0$

$$\Leftrightarrow \frac{d\left[A_1 \cdot \exp\left(\frac{-E_1}{R.T}\right) \cdot C_1 - A_2 \cdot \exp\left(\frac{-E_2}{R.T}\right) \cdot C_Q \cdot C_S\right]}{dT} = 0$$

$$\Leftrightarrow \frac{d\left[A_1 \cdot \exp\left(\frac{-E_1}{RT}\right) \cdot C_1\right]}{dT} + \frac{d\left[-A_2 \cdot \exp\left(\frac{-E_2}{R.T}\right) \cdot C_Q \cdot C_S\right]}{dT} = 0$$

$$\Leftrightarrow A_1 C_1 \exp\left(\frac{-E_1}{RT}\right) \cdot \left(\frac{-E_1}{R}\right) \cdot (-1) \cdot \frac{1}{T^2} - A_2 \cdot C_Q \cdot C_S \cdot \exp\left(\frac{-E_2}{RT}\right) \cdot \left(\frac{-E_2}{R}\right) \cdot (-1) \cdot \frac{1}{T^2} = 0$$

$$\Leftrightarrow A_1 \cdot C_1 \cdot \exp\left(\frac{-E_1}{RT}\right) \cdot \frac{E_1}{R.T^2} - A_2 \cdot C_Q \cdot C_S \cdot \exp\left(\frac{-E_2}{RT}\right) \cdot \frac{E_2}{RT^2} = 0$$

From the advancement table, we have :

	A	$\rightleftharpoons$	Q	F
i	C_{A_0}		0	0
f	$C_A = C_{A_0} \cdot (1 - X_A)$		$C_Q = C_{A_0} \cdot X_A$	$C_F = C_{A_0} \cdot X_A$

$$\Leftrightarrow A_1 \cdot C_{A_0} \cdot (1 - X_A) \cdot \exp\left(\frac{-E_1}{RT}\right) \cdot \frac{E_1}{RT^2} - A_2 \cdot (C_{A_0} \cdot X_A)^2 \cdot \exp\left(\frac{-E_2}{RT}\right) \cdot \frac{E_2}{RT^2} = 0$$

$$\Leftrightarrow \exp\left(\frac{-E_1}{RT}\right) \cdot A_1 \cdot C_{A_0} \cdot (1 - X_A) \cdot \frac{E_1}{RT^2} = A_2 \cdot (C_{A_0} \cdot X_A)^2 \cdot \frac{E_2}{RT^2} \cdot \exp\left(\frac{-E_2}{RT}\right)$$

$$\Leftrightarrow \frac{-E_1}{RT} + \ln\left[A_1 \cdot C_{A_0} \cdot (1 - X_A) \cdot \frac{E_1}{RT^2}\right] = \frac{-E_2}{RT} + \ln\left[A_2 \cdot (C_{A_0} \cdot X_A)^2 \cdot \frac{E_2}{RT^2}\right]$$

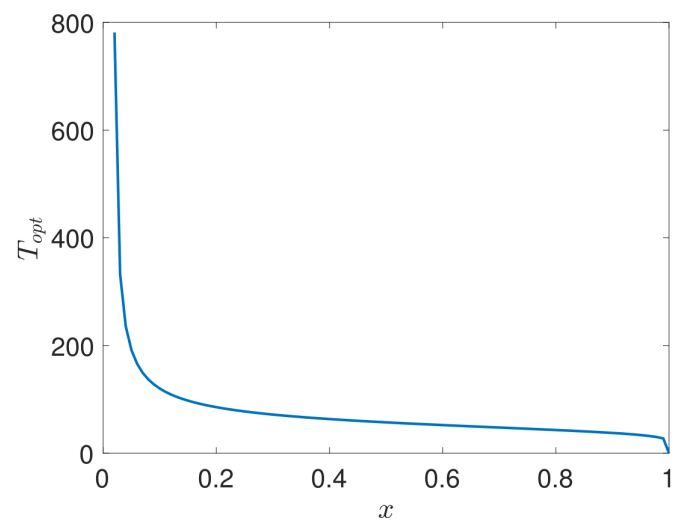
$$\Leftrightarrow \ln\left[A_1 \cdot C_{A_0} \cdot (1 - X_A) \cdot \frac{E_1}{RT^2}\right] - \ln\left[A_2 \cdot (C_{A_0} \cdot X_A)^2 \cdot \frac{E_2}{RT^2}\right] = \frac{E_1 - E_2}{RT}$$

$$\Leftrightarrow \quad \ln\left(\frac{A_1 \cdot C_{A_0} \cdot (1-X_A) \cdot \frac{E_1}{R T^2}}{A_2 \cdot (C_{A_0} \cdot X_A)^2 \cdot \frac{E_2}{R T^2}}\right) = \frac{E_1 - E_2}{R T}$$

$$\Leftrightarrow \quad \ln\left(\frac{A_1 \cdot (1-X_A) \cdot E_1}{A_2 \cdot C_{A_0} \cdot X_A^2 \cdot E_2}\right) = \frac{E_1 - E_2}{R T}$$

$$\Leftrightarrow \quad T = \frac{E_1 - E_2}{R \cdot \ln\left(\frac{A_1 \cdot (1-X_A) \cdot E_1}{A_2 \cdot C_{A_0} \cdot X_A^2 \cdot E_2}\right)}$$

$$\Rightarrow \quad T_{opt}(X_A) = \left[\frac{R}{E_1 - E_2} \cdot \ln\left(\frac{A_1 \cdot (1-X_A) \cdot E_1}{A_2 \cdot C_{A_0} \cdot X_A^2 \cdot E_2}\right)\right]^{-1}$$



EXERCISE 4

For Exercise 4

Unfortunately, the intermediate calculations are not provided in the official solution, but if you follow the steps I described you can fully reconstruct the reasoning behind the graphs and obtain the same results yourself.

In Exercise 3 we already derived the optimal temperature profile $T_{opt}(X)$, so the first question of Exercise 4 is essentially completed.

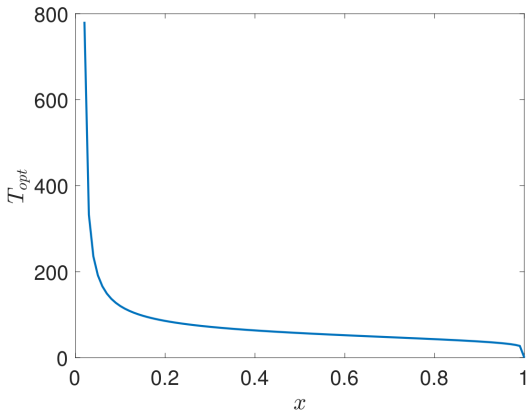
For Question 2, you now impose the maximum allowable temperature of 800 K.
At low conversion, the theoretical optimal temperature would be above 800 K, so the constrained optimal policy is to start at the maximum temperature (800 K). You keep the reactor at 800 K until the conversion reaches the point where the unconstrained $T_{opt}(X)$drops below 800 K. Beyond that conversion, you simply follow the decreasing $T_{opt}(X)$curve.

For Question 3, you integrate the continuity equation using the temperature profile from Question 2. This gives you the time evolution of both conversion and temperature.

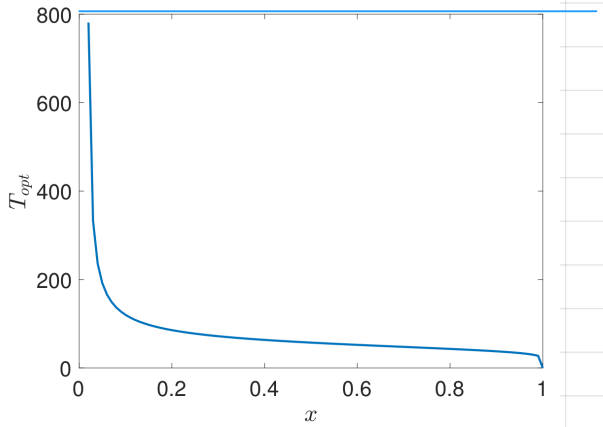
For Question 4, because the reactor is jacketed and the temperature must follow the prescribed $T(t)$, you write the energy balance to determine the heat-removal requirement :$\frac{q}{V} = (-\Delta H)\, r_A - \rho c_p \frac{dT}{dt}$. Evaluating this expression as a function of conversion gives the required heat-removal rate per unit volume.

1. Determine the optimal temperature profile as a function of conversion, in the absence of any other constraint, in order to minimize the reaction time.

See exercise 3.



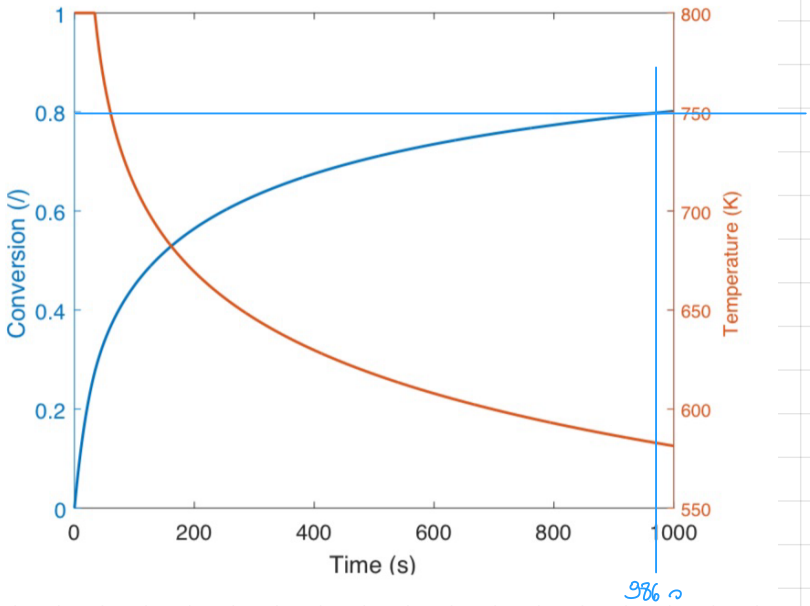
2. Modify the temperature profile obtained at previous point, knowing that the maximum allowable temperature in the reactor is 800 K, and plot it vs. conversion.



cannot go higher than this

3. Determine the evolution of conversion and temperature with time by integrating the continuity equation. In order to do so, implement a numerical routine such as Runge-Kutta, Heun or Euler method. Plot the profiles of conversion and temperature vs. time and determine the time required to reach the requested conversion.

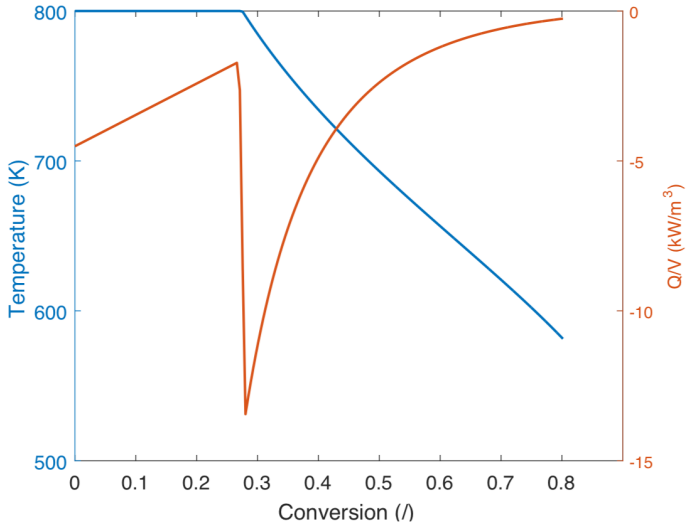
Solution: The temperature profile is calculated using the Table in the reference book (*Chemical Reactor Analysis & Design*, page 415) and then corrected to satisfy the constraint $T_{max} = 800K$. The continuity equation is integrated to calculate the conversion-versus-time profile. The conversion and corrected temperature profiles are shown versus time on the Figure below. The required time to reach a conversion of 0.8 is **986** sec, which corresponds to **16 min 26 sec**.



$\frac{q}{V}$

4. Determine the heat exchange rate per unit reactor volume required ($kJ/m_r^3.s$) and plot it vs. conversion.

The Figure below shows the corrected temperature profile and the heat exchange rate per unit reactor volume versus conversion.



$$\frac{q}{V} = (-\Delta H) \cdot r_1 - \rho C_p \frac{dT}{dt}$$

Ce que perplexity me dit mais je comprends quand même rien wesh :

For Question 4, the goal is to compute and plot the **heat-removal rate per unit reactor volume** q/V as a function of conversion, using the optimal temperature profile already obtained in Question 2–3.

1. Write the batch energy balance

Because the reactor is isothermal only by heat exchange with a jacket, the energy balance on the liquid gives the required heat-removal rate:

$$\frac{q}{V} = (-\Delta H)\, r_A - \rho c_p \frac{dT}{dt}$$

where:

- q/V is the heat exchanged with the jacket (negative when heat is removed).
- $-\Delta H$ is the exothermic reaction enthalpy (positive parameter).
- r_A is the rate of consumption of A (kmol/m^3·s).
- ρ and c_p are the mixture density and heat capacity given in the data.

This equation is the starting point for computing the required heat removal at each time during the batch.

2. Express the reaction rate r_A

For the reversible first-order reaction $A \rightleftharpoons R$:

$$r_A = -(k_1 C_A - k_2 C_R)$$

with Arrhenius kinetics

$$k_1 = A_1 \exp(-\frac{E_1}{RT}), \qquad k_2 = A_2 \exp(-\frac{E_2}{RT})$$

using the given A_1, E_1, A_2, E_2.

At each time (or conversion) you already know C_A and C_R from the batch mole balance solution of Question 3, so you can evaluate $r_A(t)$ numerically.

3. Obtain dT/dt from the prescribed temperature profile

From Question 2, an optimal temperature profile $T(X)$ respecting the 800 K constraint was derived; from Question 3, this profile was followed in time during integration of the continuity equation, giving $T(t)$.

- If $T(t)$ is analytical, differentiate it to get dT/dt.
- If $T(t)$ is only available numerically (values at discrete times), estimate dT/dt with finite differences, e.g. $(T_{i+1} - T_i)/(t_{i+1} - t_i)$ at each step.

This derivative captures how fast the temperature is changing and therefore how much sensible heat must be added or removed in addition to the reaction heat.

4. Evaluate q/V as a function of time

At every time step t_i from the numerical integration in Question 3, you now have:

- Conversion $X_i = X(t_i)$.
- Temperature $T_i = T(t_i)$.
- Reaction rate $r_{A,i}$ from the kinetics at T_i and concentrations at X_i.
- Temperature time derivative $(dT/dt)_i$.

Insert these values into the energy-balance expression

$$\left(\frac{q}{V}\right)_i = (-\Delta H) r_{A,i} - \rho c_p \left(\frac{dT}{dt}\right)_i$$

to obtain the instantaneous heat-exchange rate per unit volume at each time point.

The large negative spike in q/V near the point where the temperature hits 800 K in the final solution illustrates the strong cooling required to enforce the temperature constraint.

5. Convert from time to conversion and plot

Finally, convert the time-based results into a function of conversion:

- From the numerical solution, pair each time point with its conversion X_i.
- Use the set of pairs $(X_i, (q/V)_i)$ to plot q/V versus X.

This produces the curve shown in the solution: q/V is strongly negative around $X \approx 0.3$ where the reactor temperature is forced to stay at 800 K, then its magnitude gradually decreases and approaches zero as conversion approaches 0.8 and the temperature falls.

EXECISE 5

What reactor volume is required ?

- $\rho_{\text{total solution in}} = 1020 \ kg/m^3$
- 46 % A , 23 % B $\rightarrow$ 100 - 46 - 23 = 31 % S
- $m_R = 50\,000 \ kg/day$
- $X_B = 0,35\ \%$

We need to find the concentration of R:

	A	B	$\rightleftharpoons$	R	S
C_{i_0} $\left[\frac{kmol}{m^3}\right]$	$0,46[\%] \cdot 1020 \left[\frac{kg}{m^3}\right] \cdot \frac{1}{46} \left[\frac{kmol}{kg}\right]$	$\frac{0,23 \cdot 1020}{60}$			$\frac{(1-0,46-0,23) \cdot 1020}{18}$
	$= 10,2 \ \frac{kmol}{m^3}$	$= 3,91 \ \frac{kmol}{m^3}$		$0 \ \frac{kmol}{m^3}$	$= 17,56 \ \frac{kmol}{m^3}$
C_i $\left[\frac{kmol}{m^3}\right]$	$= 10,2 - 3,91 \cdot X_B$	$3,91 - 3,91 \cdot X_B$ $= 3,91 \cdot (1 - X_B)$		$= 3,91 \cdot X_B$	$= 17,56 + 3,91 X_B$
C_{i_f} $(X_B = 0,35)$ $\left[\frac{kmol}{m^3}\right]$	$= 8,83$	$= 2,54$		$= 1,37$	$= 18,33$

$MW_A = 2 \cdot 12 + 5 + 16 + 1 = 46 \ g/mol = 46 \ kg/kmol$
$MW_B = 12 + 3 + 12 + 2 \cdot 16 + 1 = 60 \ g/mol$
$MW_R = 12 + 3 + 12 + 2 \cdot 16 + 2 \cdot 12 + 5 = 88 \ g/mol$
$MW_S = 2 + 16 = 18 \ g/mol$

$\rightarrow$ I produce 1,37 kmol R / m^3. I'm looking for a volume $\rightarrow$ based on the units, I have :

$$V \ [m^3] = \frac{1}{1,37} \left[\frac{m^3}{kmol_R} \text{ per batch}\right] \cdot \left[\frac{kmol_R}{kg}\right] \cdot \left[\frac{kg}{day}\right] \cdot [\text{day per batch}]$$

Then we use the continuity equation to find the required time to reach the desired conversion x B = 0,35 :

$$\frac{dX_B}{dt} = \frac{V}{N_{B_0}} \cdot r_B$$

$$\Leftrightarrow \frac{N_{B_0}}{V} \cdot \frac{1}{r_B} \, dX_B = dt$$

and the rate of reaction of B is function of the conversion of B : $r_B(X_B)$

$$\Leftrightarrow C_{B_0} \cdot \left[k \cdot \left(C_1 \cdot C_B - \frac{C_R \cdot C_S}{K}\right)\right]^{-1} dX_B = dt$$

$$\Leftrightarrow 3,91 \cdot \left[7,23 \cdot 10^{-6} \cdot \left((10,2 - 3,91 X_B) \cdot (3,91 \cdot (1 - X_B)) - \frac{3,91 X_B \cdot (17,56 + 3,91 X_B)}{2,93}\right)\right]^{-1} dX_B = dt$$

$$\Rightarrow \quad t_f = 3.91 \int_0^{x_{Bf}} \frac{dx_B}{7.23 \cdot 10^{-6} \left((10.2 - 3.91x_B)3.91(1 - x_B) - \frac{1}{2.93} 3.91 x_B (17.56 + 3.91 x_B) \right)}$$
$$\simeq 7766\, s \simeq 2h\, 10 min.$$

We need to add to this reaction time the time required for filling, emptying,...

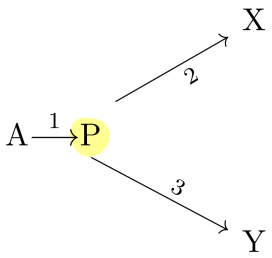
→ Total cycle time is t_c = 3 h 10 min

⇒ meaning we can perform on average $\frac{24}{3,16}$ = 7,6 cycles per day

The volume of the batch reactor is then obtained as:

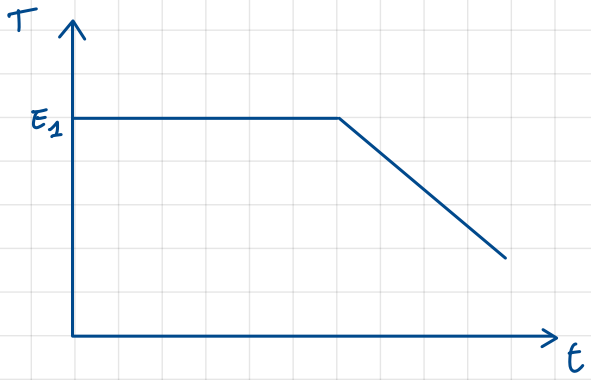
$$V_{\text{batch}} = \frac{50\,000\,[\frac{kg_R}{\text{day}}]}{7.6\,[\frac{\text{batch cycles}}{day}] \cdot 1.37\,[\frac{kmol}{m^3}\ \text{per batch}] \cdot 88\,[\frac{kg}{kmol}]} = \boxed{\simeq 55\, m^3}$$

EXERCISE 6



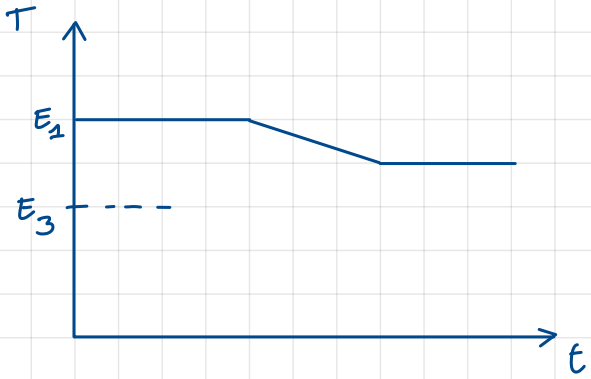
P is our product of interest, we want to avoid loosing it in X and Y.

1. $E_2 > E_3 > E_1$



In this case, the temperature should be high at the beginning in order to drive the conversion of A into P and then we should quench the reaction (sharp decrease of T) in order to avoid P lost into X and Y.

2. $E_2 > E_1 > E_3$



In this second case, the temperature should again be high at the beginning and decrease afterwards but the decrease should not be as sharp, otherwise the conversion of P to Y will be the kinetically favoured reaction and hence we will have excessive loss of P into Y.

EXERCISE 7

From the continuity equation (mass balance) we have :

$$\frac{dx_A}{dt} = \frac{V}{N_{A_0}} r_A(x_A, T)$$

$\Leftrightarrow \frac{dx_A}{dt} = \frac{V}{N_{A_0}} \cdot k \cdot C_A^n$ where n is the order of the reaction.

Knowing that $r_A = \left[\frac{kmol}{m^3 . s}\right]$ and $C_A \left[\frac{kmol}{m^3}\right]$ and given the units of $k \left[\frac{m^6}{kmol_A^2 . s}\right]$ we deduce the value of n :

$\Leftrightarrow \frac{dx_A}{dt} = \frac{V}{N_{A_0}} \cdot k(T(x_A)) \cdot C_A^3(x_A)$

↓ first find an expression of $T(x_A)$

$\frac{kmol}{m^3 . s} = \frac{m^6}{kmol_A^2 . s} \cdot \left(\frac{kmol}{m^3}\right)^n \longrightarrow n = 3$

↳ third order reaction

From the species and energy continuity equations in the adiabatic case we have :

$$\begin{cases} \frac{dx_A}{d\theta} = \frac{V}{N_{A0}} r_A(x_A, T) \\ m_t c_{p,m} \frac{dT}{d\theta} = V r_A(x_A, T)(-\Delta H) + q \cdot A_R \end{cases}$$

but this term = 0 bc we are in an adiabatic case

$$\begin{cases} r_A = \frac{dx_A}{dt} \cdot \frac{N_{A_0}}{V} & (1) \\ m_T \cdot C_p \cdot \frac{dT}{dt} = V \cdot r_A \cdot (-\Delta H) & (2) \end{cases}$$

(1) → (2)

$$m_T \cdot C_p \cdot \frac{dT}{dt} = V \cdot \frac{dx_A}{dt} \cdot \frac{N_{A_0}}{V} \cdot (-\Delta H)$$

$\Leftrightarrow \int dT = \frac{N_{A_0} \cdot (-\Delta H)}{m_T \cdot C_p} \int dx_A$

$\Leftrightarrow T - T_0 = \underbrace{\frac{N_{A_0} \cdot (-\Delta H)}{m_T \cdot C_p}} \cdot (x_A - x_{A_0})$

$\triangleq \frac{1}{\lambda}$

$\Leftrightarrow T = T_0 + \frac{1}{\lambda} \cdot (x_A - x_{A_0})$

where $\frac{1}{\lambda} = \frac{10,2\ [kmol] \cdot (-104\,670) \left[\frac{kJ}{kmol}\right]}{350\ [kg] \cdot 2,47 \left[\frac{kJ}{kg . K}\right]} = -454,38\ K$

$\Leftrightarrow \lambda = -0,0022\ K^{-1}$

Now we can reintroduce it all:

$$\Leftrightarrow \frac{dx_A}{dt} = \frac{V}{N_{A0}} \cdot \exp\left(\frac{-10\,000}{RT} + 5\right) \cdot \left(C_{A0} \cdot (1-x_A)\right)^3$$

$$\Leftrightarrow \frac{dx_A}{dt} = \frac{V}{N_{A0}} \cdot \exp\left(\frac{-10\,000}{R \cdot \left(T_0 + \frac{1}{\lambda} \cdot (x_A - x_{A0})\right)} + 5\right) \cdot \left(C_{A0} \cdot (1-x_A)\right)^3$$

$$\Leftrightarrow dt = \frac{N_{A0}}{V} \cdot \left[\exp\left(\frac{-10\,000}{R \cdot \left(T_0 + \frac{1}{\lambda} \cdot (x_A - x_{A0})\right)} + 5\right)\right]^{-1} \cdot \frac{1}{C_{A0}^2 \cdot (1-x_A)^3}\, dx_A$$

$$\Leftrightarrow t = \int_{x_{A0}}^{x_f} \left[\exp\left(\frac{-10\,000}{R \cdot \left(T_0 + \frac{1}{\lambda} \cdot (x_A - x_{A0})\right)} + 5\right)\right]^{-1} \cdot \frac{1}{C_{A0}^2 \cdot (1-x_A)^3}\, dx_A$$

$$\Leftrightarrow t = \frac{1}{C_{A0}^2} \int_{x_{A0}=0,1}^{x_f=0,7} \frac{1}{\exp\left(\frac{-10\,000}{R \cdot \left(T_0 + \frac{1}{\lambda} \cdot (x_A - x_{A0})\right)} + 5\right) \cdot (1-x_A)^3}\, dx_A$$

$$\Leftrightarrow t = 32,3\ s \quad \text{(solved numerically)}$$

REM: To fit with the original exercise, the perfect gas constant must be expressed in kcal/(kmol.K), i.e. R = 1,987 kcal/(kmol.K).